

Two reaction–diffusion species driving growth and death:

is the mechanism sound as biology, as code, and as Plexus2?

tyssue_rd_ops · tyssue_ops3d · make_two_species.py · make_sculpt.py · log/okuda/{ts,tsd,sc}_* (24 preliminary runs) · 11 August 2026

Scope. This note is about one addition to the framework: a second reaction–diffusion species, and the two operators that consume it — growth inhibition and cell death. It asks whether that addition is *sound*, on three axes: as biology, as code, and as Plexus2. **It does not ask whether the mechanism achieves the campaign’s goal.** Twenty-four preliminary runs exist in `log/okuda` and they raised a specific set of defects; §6 is that list, structured, with each one’s status. A mechanism whose implementation is unverified cannot be compared with an observation at all, so the goal question is deferred until this note’s gates close.

1. The mechanism

One field cannot pattern two processes. With a single Gray–Scott pair every operator reads the same activator — growth is gated on it, death is gated on it, the purse-string selects on it — so the tissue carries exactly one map. “Where it grows” and “where it dies” are then the same region or exact complements of it, by construction, and no setting of any parameter separates them. This is a property of the representation.

Two species is two maps. Biologically this is unremarkable and well attested: a tissue carries many morphogens at once, on different length scales, read by different targets. Two are enough to state the smallest interesting case — one field says *grow here*, another says *stop* or *die here* — and that is the case this note verifies.

The two consumers are drawn from different literatures. Growth inhibition is lateral inhibition in the Gierer–Meinhardt sense: a diffusible signal that suppresses the response of the cells around a source. Patterned death is Monier *et al.* (2015), where apoptotic force drives epithelial folding in *Drosophila* — and it is the only *inward* deformation available in this vocabulary, every other operator pushing the sheet outward.

The arithmetic that fixes the representation. A dying cell leaves an epithelium by losing neighbours until it is a triangle; a triangle collapses to a point as a T2 transition, $V - 2$, $E - 3$, $F - 1$, so $\chi = V - E + F$ is unchanged and the sheet stays closed at genus 0. Collapsing a face with $k > 3$ sides would merge k vertices into one and leave a vertex of degree k , **which a trivalent sheet cannot represent.** So death cannot be one operator: it is mark, then shed, then extrude, and the shedding belongs to `reconnect_t1_3d`. That is not a hidden coupling; it is the mechanism. A cell leaves an epithelium by losing neighbours.

Two ratios fix the proportions. The species’ wavelengths must differ or the second map is redundant: measured on a static sphere, species A at $\chi = 1.3$ holds 2 spots at a spacing of 6.99 cell diameters and species B at $\chi = 0.4$ holds 15 at 3.18 — a factor of 2.2, with **A** the coarse map. χ sets this, not the diffusivity pair: at $\chi = 1.3$ with doubled diffusivities the field does not pattern at all. And the clearing time of a dying cell is fixed by two numbers and nothing else (Eq. (4)).

2. Equations and symbols

Two independent systems in one buffer. `chem` is width 4; species A occupies columns (0,1) and species B columns (2,3). Each has its own seeder, diffusion and reaction, and each operator declares which pair it owns through `chan`:

$$\frac{\partial a_X}{\partial t} = D_a^X \nabla^2 a_X + f(a_X, u_X), \quad \frac{\partial u_X}{\partial t} = D_u^X \nabla^2 u_X + g(a_X, u_X), \quad X \in \{A, B\}, \quad (1)$$

with f, g Gray–Scott. The two systems are coupled *only* through the operators that read them, never through the buffer — which is a property that must be gated (G7), not assumed.

Growth, and the zero it did not have. The standing law is purely activating,

$$\dot{v}_{\text{eq}} = \text{rate} \cdot (\rho + \text{hill}(a_A)), \quad \text{hill}(x) = \frac{x^n}{x_{\text{sw}}^n + x^n} \in [0, 1], \quad (2)$$

so the slowest a cell can grow is $\text{rate} \cdot \rho$, and $\rho > 0$ is required by premise P1 (a tissue that adds no material is not growing). The inhibitor supplies the missing zero, multiplicatively:

$$\dot{v}_{\text{eq}} = \text{rate} \cdot (\rho + \text{hill}(a_A)) \cdot (1 - \text{hill}_{\text{inh}}(\hat{a}_B)), \quad \hat{a}_B = a_B / \max_f a_B. \quad (3)$$

$\hat{a}_B$ is normalised to the inhibitor’s own maximum for the same reason a_{sw} is a fraction: a threshold stated absolutely, against a field whose scale the chemistry sets, is either always on or always off.

Death, as three steps and one closed-form time.

$$V_f^0 \leftarrow (1-s)V_f^0, \quad \text{extrude when } V_f^0 < c v_{\text{ref}}, \quad \tau_{\text{clear}} = \frac{\ln c}{\ln(1-s)} \text{ ticks} \quad (4)$$

Volume is conserved into the survivors, and the conservation is bounded. When f dies its remaining material is shared over its live ring neighbours, and the increment is capped so the recipient stays inside the integrator’s basin:

$$c_g \leftarrow c_g + \min\left(\frac{\text{amt}/|\mathcal{L}|}{V_g^0}, c_{\text{max}} - c_g\right), \quad \mathcal{L} = \{g \in \mathcal{N}(f) : V_g^0 \geq c v_{\text{ref}}\}, \quad (5)$$

with c_{max} the field’s own running maximum. Material that will not fit is *dropped and counted* in `apop_spill`, so the conservation law reports its own error rather than violating itself silently. Both restrictions are gated (G4, G5).

A death RATE, not a death set: with N live cells, $|\{\text{marked}\}| \leq \phi N$, and when the cap binds the worst candidates are taken by the mode’s own severity. Geometric modes are exempt, because a set chosen once has no flux.

a_X, u_X	activator, substrate of species X	<i>per cell</i> , INTENSIVE. A concentration, so it is meaningful to compare between cells of different size
<code>chan</code>	species selector	which <code>chem</code> pair an operator owns. Default 0, so every pre-existing spec is unchanged
D_a^X, D_u^X	diffusivities	the <code>RATIO</code> sets the Turing wavelength; the pair sets the speed
ρ	growth baseline	tip:body is $(\rho + 1) : \rho$. $\rho = 0$ is an infinite ratio, which P1 refuses
x_{sw}, n	gate threshold, sharpness	a <code>FRACTION</code> of that field’s own maximum, never absolute
s, c	<code>shrink_rate</code> , <code>critical_frac</code>	per tick, and a fraction OF v_{ref}
v_{ref}	reference volume	the <code>SEED-TIME</code> median. Never recomputed: recomputing it after deaths moves every threshold in the model
τ_{clear}	clearing time	INTENSIVE: a property of (s, c) alone, not of the tissue
ϕ	<code>max_mark_frac</code>	fraction of LIVE cells under sentence at once — bounds a FLUX
n_{apop}	deaths	EXTENSIVE: scales with the tissue, so never comparable between runs of different size without dividing by N
<code>apop_spill</code>	unplaced material	EXTENSIVE. The conservation law’s own error term

3. The Plexus2 container

No new set. What is new is a *family* the vesicle lacked — *Die* — and a second species inside one state buffer. Three claims about the container have to hold, and each is a gate:

1. **Two instances of one operator are ADDITIVE, not overwriting.** `cell_react` returns a delta over the whole `chem` tensor and writes zeros outside the pair its `chan` names. Without that, a second instance overwrites the first and “two species” is a name for one. (G7)
2. **apoptosis_3d is structural, not lateral.** It changes how many entities exist, so it may not be a pure state map: it rennumbers faces, and every per-face array and every *pending delta* must be permuted with it. (G3)
3. **A named population is a seed window, one level up.** `list`, `band` and `cone` choose a population ONCE. Re-evaluated every frame, a death sentence becomes a death rate — the same argument the paper makes for seeding, where an initial condition re-applied every step is a moving boundary condition. (G6)

operator	kind / family	reads	mechanism	gate
<code>seed_cell_rd</code>	seed / growth	—	seeds its own columns	G7
<code>cell_diffuse</code>	lateral / fields	<code>chem</code> , adjacency	per-column diffusivity vector	G7
<code>cell_react</code>	lateral / fields	<code>chem</code>	Eq. (1), zeros outside its pair	G7
<code>grow_3d</code>	structural / growth	<code>chem</code>	Eq. (3)	G8, G9
<code>apoptosis_3d</code>	structural / die	<code>chem</code> , mesh	Eqs. (4)–(5)	G1–G6
<code>reconnect_t1_3d</code>	rewire / topology	mesh	sheds neighbours — REQUIRED	G2

Schedule. Death runs *between growth and the mechanics*: growth sets the targets, death overrides them for the sentenced, and the relaxation sees both in one tick, so a cell extruded this frame has its hole closed this frame. Measured, same composition with death appended after the recorder instead: grip 0.031 against 0.110. There is no `substep_dt` block — this vesicle integrates at one rate and `shape_energy_3d` carries `relax_iters` internally.

4. The gates

Thresholds set before the run. The tiers are the paper’s: **bookkeeping** asks whether the code does what the operator says, **closed form** whether it reproduces physics it was given, and **measurement** whether it agrees with something observed in cells.

gate	threshold, before	measured
<i>tier 1 — bookkeeping</i>		
G1 every selector fires on a known population and NOT on a uniform field	all cases	19/19
G2 $\chi = 2$ at every frame with death running	exact	2, all frames
G3 a topology change permutes pending DELTAS, not only state	activator ≥ 0	0 (was -1.04×10^{10})
G4 a bequest goes only to neighbours above the extrusion threshold	no divisor $< c v_{\text{ref}}$	P4, P12 cleared
G5 a bequest leaves the recipient inside the integrator’s basin	no NaN, activator ≥ 0	RE-RUNNING —
G6 a named population is chosen ONCE; only state modes re-evaluate	one death per named cell	1, was 7
G7 two species never write each other’s columns	$\{0, 1\}$ and $\{2, 3\}$	exact, non-uniform
G8 inhib_chan absent reproduces a pre-change run	metrics identical	RUNNING — §6 I
G9 growth $\rightarrow 0$ as the inhibitor saturates	$< 5\%$ of uninhibited	$0.978 \rightarrow 0.014$
G10 the cap admits at most ϕN , worst first, and drains	exact	8/100; took worst;
G11 n_{apop} equals the cells removed	exact, death-only composition	$400 - N_{\text{final}}$
<i>tier 2 — closed form</i>		
G12 a collapse is a T2: $V - 2, E - 3, F - 1$	exact, else refuse	refuses $k > 3$, re-ch
G13 clearing time equals $\ln c / \ln(1 - s)$	within 1 tick	NOT MEASURED
G14 the throughput lever is s , not ϕ	deaths $\propto 1/\tau_{\text{clear}}$	$\phi \times 50 \rightarrow \times 1.7; s \times 3 \rightarrow$
G15 a geometric chisel removes exactly its declared population	exact	5/5 geometries
G16 the second species’ wavelength differs from the first	spot count differs	NOT MEASURED
<i>tier 3 — measurement</i>		
G17 apoptotic rate matches an epithelium’s	1–3% of cells per day	BLOCKED: no unit
G18 clearance time matches an extrusion in vivo	30–90 min	BLOCKED: no unit
G19 the two species’ wavelengths bracket a real morphogen pair	—	BLOCKED: no unit

Eleven bookkeeping, five closed form, three measurement — and all three of the last are blocked for want of a **units**: block, the same defect the **ecm** study found when “0.82 grid cells” turned out to be $15 \mu\text{m}$. So this note can establish that the operators do what they say and reproduce the arithmetic they were given. **It cannot yet say anything about cells.** Declaring units is the single change that would move three gates from blocked to failable.

5. What the runs establish

Twenty-four exploratory runs on **r001_03** and fifteen dedicated gate runs. Three results are properties of the operators rather than of any morphology, and are the ones this note rests on.

Position dominates rate. Death patterned in **SPACE** deforms the body at a thirtieth of the cell cost of death patterned by rate:

run	deaths	cells	protr_f	invag_f	red_vol
sc_crown (a polar cap)	113	16056	1.267	0.343	0.814
sc_waist (an equatorial band)	332	14486	1.174	0.328	0.807
tsd_max (the highest rate)	3460	2895	1.140	0.111	0.957
control (no death)	0	12692	1.095	0.120	0.959

Scattered death thins a ball into a smaller ball; a chosen population is a cut.

The throughput lever is τ_{clear} , not the cap. Fifty times ϕ moves the death count by $1.7\times$; tripling s at fixed ϕ moves it by $4.7\times$. The queue does not fill, so what binds is how fast a sentenced cell reaches the extrusion threshold (G14).

Inhibition is monotone and is a brake. $12692 \rightarrow 6920 \rightarrow 4859 \rightarrow 4154$ cells as **inhib_sw** tightens, with **protr_final** falling $1.095 \rightarrow 1.020$. Eq. (3) asserts a multiplicative shutdown and the tissue size follows it; the shape does not sharpen.

6. Gate outcomes and open items

G8 — the engine is deterministic, and the trajectory still moved. Two replicates of one spec, same seed, different names: **g8_rep1** and **g8_rep2** agree to every recorded digit (7099 cells, **protr_peak** 1.175, **act_max**

0.825). So a per-run comparison inside one build is exact, and the seed spread is the only noise floor within it. But both differ from the same spec’s archived run (7261 cells, 1.176): a numerically equivalent refactor of `cell_diffuse`’s coefficient vector and `cell_react`’s delta shifted a chaotic trajectory by 2.2% in cell count over 1800 frames. **Comparisons ACROSS that change are not exact; comparisons within a build are.**

G13 — consistent, and not resolvable at this stride. One named cell on a static sheet, four shrink rates. The death frames order as predicted ($\sim 150, 100, 100, 75$ against 87, 68, 62, 58) and each exceeds its prediction, which is the shedding time Eq. (4) does not include. But `metrics.npz` carries 25 samples for 601 frames — a stride of 25 — and the differences between rungs are of the same order. **The formula is not contradicted and is not confirmed.** Closing it needs the frame each marked cell crosses cv_{ref} recorded beside the frame it is extruded, which separates the two terms and does not depend on the sampling stride at all.

G16 — closed, failed, and the direction is the finding. Species B is the FINE map: 15 domains at a spacing of 3.18 cell diameters against A’s 2 at 6.99. χ sets this, not the diffusivity pair — at $\chi = 1.3$ with doubled diffusivities the field does not pattern at all (`act_max` 0.0). The requirement the gate stands for, that the two maps differ, holds at $2.2\times$.

G5 — closed on the run it was built for. `tsd_cap10` previously reached `act_mean_floor` -7.29×10^{11} and went NaN at frame 1430. With the bequest bounded per column it finishes with `act_max_final` 0.579, `act_mean_floor` 0.0465, no extinction and no premise broken — and `apop_spill` 3.009, so material WAS refused and the bound engaged rather than passing vacuously. That also closes **I8**: the conservation law’s error term is recorded and readable.

I4 — closed, and it confirms the throughput law while removing the confound. The slow ladder with `inhib_chan` absent:

run	τ_{clear}	deaths	cells	<code>apop_spill</code>
i4_slow005	37	875	8482	4.14
i4_slow002	94	729	10534	2.45
i4_slow001	189	199	11553	0.29

Deaths fall and tissue rises monotonically with the clearing time, which is Eq. (4) acting as the throughput lever (G14) measured directly. And the confounded series’ small tissue (3525–4353 cells) was the inhibitor, not the death speed: at the same shrink rates without it the tissue holds 8482–11553.

	item	what closing it requires	state
G5	the conservation bound holds	closed on <code>tsd_cap10</code>: finite activator, no premise broken, <code>apop_spill</code> 3.009. <code>tsd_cap25</code> confirms it at the higher cap.	<code>cap25</code> running
G8	runs are reproducible	deterministic within a build; not across the channel refactor. Any comparison spanning it must be re-measured.	qualified
G13	$\tau_{\text{clear}} = \ln c / \ln(1 - s)$	a per-cell record of the crit-crossing frame, independent of the sampling stride.	open
G16	the two maps differ	closed: $2.2\times$ in spacing, with A coarse and B fine.	done
G16c	the pair keeps both wavelengths	g16c_pair re-measured. The per-species pass was first added to <code>measure_1frame</code> , which <code>run_one</code> does not call — it carries its own per-frame loop — so the instrument is now in that loop and the frame tuple carries channel 2.	running
I4	the slow ladder, unconfounded	closed: deaths 875 $\rightarrow$ 729 $\rightarrow$ 199 and cells 8482 $\rightarrow$ 11553 as τ_{clear} rises 37 $\rightarrow$ 189.	done
I8	<code>apop_spill</code> is readable	closed: 0.29–4.14 across the I4 ladder, rising with the death count, so the term tracks the refusals rather than sitting at zero.	done
G17–19	any measurement-tier gate	a units : block.	blocked

The instrument for the second species now exists. `b_act_max`, `b_act_cv`, `b_n_spots`, `b_spot_spacing_cells` and `b_spot_frac` are the channel-2 twins of the metrics that read channel 0, each declaring **requires** so they are UNDEFINED — not zero — when the second species is extinct. They are registered as diagnostics rather than admitted: a metric becomes admissible when a prediction may rest on it, and that requires it to reach a real run’s summary, which no two-species run has done since they were added.

7. What this licenses, and what it does not

Licensed. Two RD species coexist in one buffer without leaking (G7), which is the structural claim the whole addition rests on. The Die family is sound as bookkeeping: selectors fire where they should and nowhere else (G1), the sheet stays closed (G2), a topology change carries its pending deltas (G3), the count is exact (G11), the rate is a rate (G10), and a named population is a seed window rather than a per-frame rule (G6). Growth can be switched off (G9), and the collapse is a genuine T2 (G12).

Not licensed. **G5** is the one that matters most: until it closes, the operator can drive the chemistry out of its basin and a run's numbers are then about an integrator. **G13** means the clearing time — the quantity every death experiment is denominated in — is asserted and not measured. **No gate is a measurement** (G17–G19 blocked for want of a `units:` block), so nothing here has been compared with a cell.

The mechanism should not enter the loop until G5, G8 and G13 close. G5 because a mechanism that can NaN the chemistry will produce runs the campaign records as evidence; G8 because an operator that changes runs which do not use it corrupts every comparison in the ledger; G13 because the clearing time is the quantity every death experiment is denominated in. G16 and I4/I5 are about the experiments, not the operators, and can close afterwards.

8. References

References

- [1] Monier, B., Gettings, M., Gay, G., Mangeat, T., Schott, S., Guarner, A., Suzanne, M. (2015) Apico-basal forces exerted by apoptotic cells drive epithelium folding. *Nature* **518**:245–248.
- [2] Gierer, A., Meinhardt, H. (1972) A theory of biological pattern formation. *Kybernetik* **12**:30–39. — lateral inhibition, of which Eq. (3) is a coarse form.
- [3] Gray, P., Scott, S. K. (1984) Autocatalytic reactions in the isothermal, continuous stirred tank reactor. *Chem. Eng. Sci.* **39**:1087–1097. — both species.
- [4] Okuda, S., Inoue, Y., Eiraku, M., Sasai, Y., Adachi, T. (2013) Reversible network reconnection model for simulating large deformation in dynamic tissue morphogenesis. *Biomech. Model. Mechanobiol.* **12**(4):627–644. — the T1/T2 machinery of §1.
- [5] Turing, A. M. (1952) The chemical basis of morphogenesis. *Phil. Trans. R. Soc. B* **237**:37–72. — why two species with different diffusivities is the minimal statement.